How intrusions compromise your system

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In this section of the course, you learned that every network has inherent vulnerabilities and could become the target of a network attack.

Attackers could have varying motivations for attacking your organization's network. They may have financial, personal, or political motivations, or they may be a disgruntled employee or an activist who disagrees with the company's values and wants to harm an organization's operations. Malicious actors can target any network. Security analysts must be constantly alert to potential vulnerabilities in their organization's network and take quick action to mitigate them.

In this reading, you'll learn about network interception attacks and backdoor attacks, and the possible impacts these attacks could have on an organization.

Network interception attacks

Network interception attacks work by intercepting network traffic and stealing valuable information or interfering with the transmission in some way.

Malicious actors can use hardware or software tools to capture and inspect data in transit. This is referred to as **packet sniffing**. In addition to seeing information that they are not entitled to, malicious actors can also intercept network traffic and alter it. These attacks can cause damage to an organization's network by inserting malicious code modifications or altering the message and interrupting network operations. For example, an attacker can intercept a bank transfer and change the account receiving the funds to one that the attacker controls.

Later in this course you will learn more about malicious packet sniffing, and other types of network interception attacks: on-path attacks and replay attacks.

Backdoor attacks

A **backdoor attack** is another type of attack you will need to be aware of as a security analyst. An organization may have a lot of security measures in place, including cameras, biometric scans and access codes, to ensure employees do not enter and exit unseen. However, an employee might work around the security measures by finding a backdoor to the building that is not as heavily monitored, allowing them to sneak out for the afternoon without being seen.

In cybersecurity, backdoors are weaknesses intentionally left by programmers or system and network administrators that bypass normal access control mechanisms. Backdoors are intended to help programmers conduct troubleshooting or administrative tasks. However, backdoors can also be installed by attackers after they've compromised an organization to ensure they have persistent access.

Once the hacker has entered an insecure network through a backdoor, they can cause extensive damage: installing malware, performing a denial of service (DoS) attack, stealing private information or changing other security settings that leaves the system vulnerable to other attacks. A **DoS attack** is an attack that targets a network or server and floods it with network traffic.

Possible impacts on an organization

As you've learned already, network attacks can have a significant negative impact on an organization. Let's examine some potential consequences.

- **Financial**: When a system is taken offline with a DoS attack or some other tactic, they prevent a company from performing tasks that generate revenue. Depending on the size of an organization, interrupted operations can cost millions of dollars. Reparation costs to rebuild software infrastructure and to pay large sums associated with potential ransomware can be financially difficult. In addition, if a malicious actor gets access to the personal information of the company's clients or customers, the company may face heavy litigation and settlement costs if customers seek legal recourse.
- **Reputation**: Attacks can also have a negative impact on the reputation of an organization. If it becomes public knowledge that a company has experienced a cyber attack, the public may become concerned about the security practices of the organization. They may stop trusting the company with their personal information and choose a competitor to fulfill their needs.
- **Public safety**: If an attack occurs on a government network, this can potentially impact the safety and welfare of the citizens of a country. In recent years, defense agencies across the globe are investing heavily in combating cyber warfare tactics. If a malicious actor gained access to a power grid, a public water system, or even a military defense communication system, the public could face physical harm due to a network intrusion attack.

Key takeaways

Malicious actors are constantly looking for ways to exploit systems. They learn about new vulnerabilities as they arise and attempt to exploit every vulnerability in a system. Attackers leverage backdoor attack methods and network interception attacks to gain access to sensitive information they can use to exploit an organization or cause serious damage. These types of attacks can impact an organization financially, damage its reputation, and potentially put the public in danger. It is important that security analysts stay educated in order to maintain network safety and reduce the likelihood and impact of these types of attacks. Securing networks has never been more important.

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Give me a summary

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This reading focuses on common network attack types, their mechanisms, and the potential consequences for organizations.

Network Attack Vulnerabilities

- Every network has inherent vulnerabilities that can be exploited by attackers with various motivations, including financial gain, personal reasons, political agendas, or disgruntled employment.
- Security analysts must remain vigilant to identify and mitigate potential vulnerabilities in their organization's network.

Types of Network Attacks

- **Network Interception Attacks:** These involve intercepting network traffic to steal information or interfere with data transmission, often through "packet sniffing" to capture and inspect data in transit. Attackers can also alter intercepted traffic, such as changing bank transfer details.
- **Backdoor Attacks:** These exploit intentional weaknesses left by programmers or administrators to bypass normal access controls, or backdoors installed by attackers for persistent access after an initial compromise. Once inside, attackers can install malware, launch Denial of Service (DoS) attacks, steal private information, or alter security settings.

Impacts of Network Attacks

- **Financial Impact:** Attacks can lead to significant financial losses due to interrupted operations, reparation costs, ransomware payments, and potential litigation from compromised customer data.
- **Reputational Damage and Public Safety Risks:** Cyberattacks can damage an organization's reputation, leading to a loss of public trust and customers. In critical infrastructure like government networks, power grids, or public water systems, attacks can pose serious risks to public safety and welfare.

And if you want to continue exploring this topic, try one of these follow-up questions:

- What is a backdoor attack in network security?
- How can you detect and prevent network interception attacks?
- How might a backdoor attack impact an organization's overall security posture?

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